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LETTERS TO THE EDITOR

REGRESSION TO THE MEAN AND CROSSOVER TRIALS REVISITED

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I had hoped that my comments in the pages of this journal[1] would clear up the position as regards regression to the mean and crossover trials.[2] From the reply which Grender *et al.* have made to my letter[3] it seems, however, that there is still some residual confusion. I hope to clear the matter up now.

Grender *et al.*[3] concede that several of the formulae in their paper were wrong and that one of the consequences of this mistake is that the variance of the restricted estimate of the treatment contrast, for the example they considered, is not 13·72 for an untruncated sample but 3·43. They still maintain, however, that the variance of the corresponding estimate from a truncated sample is 3.78. Curiously, they see no need to alter their major conclusions despite the fact that they appear to believe there is a slight advantage to the untruncated sample rather than a massive disadvantage.

Grender *et al.* state, 'In reference to the variances of truncated and untruncated estimates it can be seen from the equations in Table VII that they are not the same.'[3] (p. 1088) This statement is peculiar. In view of the gross error in the final column of this table which I uncovered and which they grant exists, it is inappropriate that they should rely on this incorrect table for authority. In stating that the variances of estimates for treatment contrasts from truncated and untruncated samples were identical I was claiming, of course, that the formulae for variances for the truncated sample were also wrong. In fact, even the most cursory examination of these formulae shows that they must be incorrect. For example, in their original paper[3] they have as the formula for the variance of a treatment contrast from a trunated sample:

$$(a/4)[2\sigma^2(bd + 1 - \rho^2) - 2d]\,. \tag{1}$$

However, since $a = (1/n_1 + 1/n_2)$ and $d = \rho(1 + \rho)$, it is clear that (1) is not even a multiple of σ^2 and so does not satisfy dimensional analysis and *must* be wrong. Similarly, many of the formulae for variances in the table in their reply to my letter are not multiples of σ^2 and so must also be wrong.

As I said in my original letter,[1] given the simplest correlation structure considered by Grender *et al.*,[2] it makes no difference to the variance of treatment contrasts whether or not the sample is truncated. Since my argument was not accepted I shall now amplify it.

Let Y_{ij} be a random variable denoting the jth observation $j = 1$ to 4 on the ith patient. Y_{i1} and Y_{i3} correspond to first and second baselines and Y_{i2} and Y_{i4} to measurements at the end of the first and second treatment period. Suppose

$$Y_{ij} = E[Y_{ij}] + \psi_i + \varepsilon_{ij} \tag{2}$$

where $E[\psi_i] = E[\varepsilon_{ij}] = 0$ and where the values of the $E[Y_{ij}]$ depend upon the parameters of the model and vary according to the value of j and the treatment sequence to which the ith patient has been allocated.

Let $E[\psi_i^2] = \sigma_b^2$ for all i and $E[\varepsilon_{ij}^2] = \sigma_w^2$ for all i and for all j and suppose that the ε_{ij} are independent of each other and of the ψ_i.

The is the simple standard model for errors in a crossover design with ψ_i as a 'between patient' error and ε_{ij} as a 'within-patient' error.

If we define $\sigma^2 = \sigma_b^2 + \sigma_w^2$ and $\rho = \sigma_b^2/\sigma^2$ so that $\sigma_w^2 = (1 - \rho)\sigma^2$ then we have $\text{var}(Y_{ij}) = \sigma^2$ and $\text{cov}(Y_{ij}\, Y_{i'j'}) = 0,\ i \neq i'$ and $\text{cov}(Y_{ij}\, Y_{i'j'}) = \rho,\ i = i',\ j \neq j'$. This is equivalent to the simple covariance structure considered by Grender *et al.*[2]

Now, the restricted estimate of the treatment contrast considered by Grender *et al.* is a simple linear combination of the differences:

$$Y_{i4} - Y_{i2} = E[Y_{i4}] - E[Y_{i2}] + \varepsilon_{i4} - \varepsilon_{i2} \tag{3}$$

which by inspection and under the model is obviously independent of Y_{i1}. It thus follows that the variance of (3) is the same whatever the value of the first baseline. Consequently, if this simple correlation structure applies, whether or not patients are selected according to their baseline is irrelevant and variances from truncated and untruncated estimates are identical.

The argument above is (essentially) the argument I outlined in verbal form in my original letter.[1] If it is not found convincing an alternative argument is possible in terms of the moments of the truncated trivariate normal distribution.

We may simplify this argument without affecting its essence by supposing that $E[Y_{i1}] = E[Y_{i2}] = E[Y_{i4}] = \mu$. Let k be the truncation point as defined in Grender *et al.*[2] and, following their notation, let $z = (k - \mu)/\sigma$, $t = \phi(z)/(1 - \Phi(z))$, where $\phi(z)$ is the ordinate of the standard normal distribution and $\Phi(z)$ is its cumulative probability density, and let $b = t(z - t)$.

The ordinary regressions for Y_{i2} and Y_{i4} on Y_{i1} are:

$$E[Y_{i2} | Y_{i1} = y_{i1}] = E[Y_{i4} | Y_{i1} = y_{i1}] = \mu + \rho(y_{i1} - \mu). \tag{4}$$

The conditional variances of Y_{i2} and Y_{i4} given Y_{i1} are:

$$V[Y_{i2} | Y_{i1} = y_{i1}] = V[Y_{i2} | Y_{i1} = y_{i1}] = (1 - \rho^2)\sigma^2 \tag{5}$$

Finally, as follows from the standard formula for partial correlation, the conditional covariance of Y_{i2} and Y_{i4} given Y_{i1} is:

$$\operatorname{cov}[Y_{i2} Y_{i4} | Y_{i1} = y_{i1}] = \rho(1 - \rho)\sigma^2 \tag{6}$$

Suppose Y_{i1}, Y_{i2}, and Y_{i4} have a multivariate normal distribution then standard theory of the truncated normal[4] shows that the variance of Y_{i1} conditional on Y_{i1}'s being greater than k is

$$V[Y_{i1} | Y_{i1} > k] = \sigma^2(b + 1)\,. \tag{7}$$

Putting together (4), (5), (6) and (7) one may then show that

$$V[Y_{i2} | Y_{i1} > k] = V[Y_{i4} | Y_{i1} > k] = \rho^2\sigma^2(b + 1) + (1 - \rho^2)\sigma^2 \tag{8}$$

and

$$\operatorname{cov}[Y_{i2} Y_{i4} | Y_{i1} > k] = \rho^2\sigma^2(b + 1) + \rho(1 - \rho)\sigma^2. \tag{9}$$

Now from the standard formula for the variance of the difference of two random variables and using (8) and (9) we find

$$V[Y_{i4} - Y_{i2} | Y_{i1} > k] = 2(1 - \rho^2)\sigma^2 - 2\rho(1 - \rho)\sigma^2 = 2(1 - \rho)\sigma^2. \tag{10}$$

However, (10) is the same as the formula for the unconditional variance of $Y_{i4} - Y_{i2}$ so that

$$V[Y_{i4} - Y_{i2} | Y_{i1} > k] = V[Y_{i4} - Y_{i2}] \tag{11}$$

and since the restricted estimated of the treatment contrast is simply a linear combination of these pairwise within-patient differences it follows that whether or not patients are selected on the basis of their baseline values makes not the slightest difference to the variance of the treatment estimate. Hence the variances of restricted estimates of the treatment contrasts are identically $a\sigma^2(1 - \rho)/2$ whether obtained from truncated or untruncated samples.

This equality of variances is conditional on the number of patients studied on each sequence being identical in the two cases. As I stated in my letter, it will in practice be easier to obtain the necessary numbers in the untruncated case. Grender *et al.*'s comments regarding hospital patients do not affect my opinion regarding the validity of this observation.

The paper by Grender *et al.* merely serves to confirm my general opinion regarding the use of baselines to model carryover in crossover in crossover trials: *no good ever comes of it*. The position as I see it is very simple. Either we are concerned that the baselines may be affected by carryover in which case it is best to discard them altogether and use the outcomes only (the second outcome is further from the previous treatment than is the corresponding baseline and so is less likely to be affected by carryover), or we believe that the baselines should be unaffected by carryover in which case we may use them as covariates (but for this to be useful a more complicated correlation structure than the simplest structure considered by Grender *et al.* must apply; see Senn,[5] pp. 54–60 for a discussion of the use of baselines in the AB/BA crossover.) Such an analysis deals with any chance imbalance in baselines – a point I made in my original letter and which Grender *et al.* have either misunderstood or misinterpreted describing my argument as 'statistical sophistry'[3] (p. 1089).

There is no sophistry involved; the issue of balance, conditioning and inference is subtle but fundamental. A factor is validly allowed for if we condition on it. This does not require that the factor be balanced. Even if the factor is unbalanced we can provide a valid estimate if we condition on it. The procedure is used commonly in crossover trials (where the number of patients in a given trial may not be equal on each sequence) to produce treatment estimates which are unbiased by any period effect. Such conditioning will lead to more efficient estimates if we do in fact have balance but even if we have balance, ignoring a factor is not a valid option – else we could analyse a matched pairs design as if it were an independent samples design. Hence balance is a question of efficiency; validity is a matter of conditioning.

Of course, we prefer efficient estimators. In planning a crossover trial and specifying a sample size we make assumptions about the within-patient variance and about the number of drop-outs etc. If we are going to use analysis of covariance we also have to make assumptions about the degree of correlation between baseline and outcome and the balance or otherwise of the baselines. All such assumptions may be disappointed in practice but if we randomize, the degree of baseline balance is actually one factor we can most reasonably make allowances for.

Finally, I think it is appropriate to state my opinion regarding the practical importance of all this. The crossover trial has had a bad press recently (largely undeservedly so) but of all the clinical trial designs it is one which is least affected by regression to the mean. It is my opinion that modelling such a phenomenon in the context of the crossover trial is a game which, even if played correctly, is unlikely to be worth the candle.

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4. Senn, S. J. 'Estimation for uncontrolled experiments subject to regression to the mean,' Ph.D. Thesis, Dundee University, 1985.
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AUTHORS' REPLY

The main point of Senn's original letter[1] appears to be that regression toward the mean is of little concern in 2 × 2 crossover designs. In that letter he states 'it is not possible for selection on baselines to affect treatment effects via regression toward the mean'. However, the final paragraph of his current letter states that 'of all the clinical trial designs it is one which is *least affected* by regression toward the mean' (our italics). Our main point is that regression toward the mean *can be* (this, of course, does not imply *always is*) an effect that should be considered in the analysis of such designs. This is not an indictment of crossover designs nor is it a claim that regression toward the mean is of greater concern in crossover designs than in other designs. As we discussed in the introduction and summary of Grender *et al.*,[2] specific criteria must be met in order for the baseline screening variable not to introduce bias in the estimates of contrasts of interest in a 2 × 2 crossover design with baseline measurements. These criteria include equivalent sequence groups (means, variances, screening values, etc.) and uniform covariance structure (constant correlation across time points).

We gave algebraic expressions for the regression effect (bias) of the usual linear functions of sample mean estimates of the relevant model effects in Tables II–IV of Grender *et al.*[2] From these tables the conditions under which the effect of regression toward the mean vanishes is clear. This regression effect need not vanish for the carryover contrasts if the mean and variance of the response are not the same in the two sequence groups, nor if the truncations on the baseline observations are unequal in the two groups. In order for the treatment contrast to experience no regression toward the mean bias, the mean, variance and cutoff point for the two groups must be equal, and the correlation structure must be uniform. Senn fails to address these points in his letter, focusing instead on the variance of the treatment contrast estimators in truncated versus untruncated samples. In his current letter, Senn maintains that the variances are the same if subjects are selected for the study conditional on their baseline observations being greater than a certain value (truncated baseline value) or if the baseline value is from an untruncated distribution.

There appears to be residual confusion from our paper and therefore we take this opportunity to clarify our position in more detail. Let $\mathbf{y}_{ik} = [y_{i1k}, y_{i2k}, y_{i3k}, y_{i4k}]'$ represent four consecutive measurements on the kth subject from the ith group; $i = 1, 2$; $k = 1, 2, \ldots, n_i$. Let $\operatorname{var}(y_{ijk}) = \varepsilon_i^2$ and $E(y_{ijk}) = \mu_i$, and let the covariance structure of the observations from group i before truncation be

$$\operatorname{cov}(\mathbf{y}_{ik}) = \varepsilon_i^2 \begin{bmatrix} 1 & \rho_{12} & \rho_{13} & \rho_{14} \\ \rho_{12} & 1 & \rho_{23} & \rho_{24} \\ \rho_{13} & \rho_{23} & 1 & \rho_{34} \\ \rho_{14} & \rho_{24} & \rho_{34} & 1 \end{bmatrix}.$$

We consider three effects associated with treatments A and B: first-order carryover, θ; second-order carryover, λ; and treatment, τ. We also considered the contrasts:

$$\Delta_1 = \theta_B - \theta_A,$$

$$\Delta_2 = \lambda_B - \lambda_A | (\theta_B - \theta_A = 0),$$

and

$$\Delta_3 = \tau_B - \tau_A | (\theta_B - \theta_A = \lambda_B - \lambda_A = 0).$$

The usual estimators of these contrasts are, respectively,

$$D_1 = (\bar{y}_{13\cdot} - \bar{y}_{11\cdot}) - (\bar{y}_{23\cdot} - \bar{y}_{21\cdot}),$$

$$D_2 = (\bar{y}_{12\cdot} - \bar{y}_{11\cdot} + \bar{y}_{14\cdot} - \bar{y}_{13\cdot}) - (\bar{y}_{22\cdot} - \bar{y}_{21\cdot} + \bar{y}_{24\cdot} - \bar{y}_{23\cdot}),$$

and

$$D_3 = \tfrac{1}{2}[\bar{y}_{14\cdot} - \bar{y}_{12\cdot}) - (\bar{y}_{24\cdot} - \bar{y}_{22\cdot})]$$

If the first baseline measurement is not truncated, these estimators have expected values of zero under the usual assumptions. Based on the truncated baseline measurement, we considered the event

$$T = \bigcap_{k=1}^{n_i} y_{i1k} > c_i.$$

If the data follow a multivariate normal distribution truncated on the baseline measurement then the estimators are biased and Table IV of Grender *et al.*[2] gives the following expected values:

$$E(D_1 | T) = (\theta_B - \theta_A) + (1 - \rho_{13})\eta$$

$$E(D_2 | T) = (\lambda_B - \lambda_A) + (1 - \rho_{12} + \rho_{13} - \rho_{14})\eta$$

$$E(D_3 | T) = (\tau_B - \tau_A) + (\rho_{14} - \rho_{12})\frac{\eta}{2},$$

where $\eta = (t_1 \varepsilon_1 - t_2 \varepsilon_2)$ and

$$t_i = \frac{\phi\left(\dfrac{c_i - \mu_i}{\varepsilon_i}\right)}{1 - \Phi\left(\dfrac{c_i - \mu_i}{\varepsilon_i}\right)}, \quad \phi(u) = \frac{1}{\sqrt{2\pi}} \exp^{-u^2/2}, \quad \Phi(u) = \int_{-\infty}^{u} \phi(t)\,dt.$$

Table I. Variance of restricted contrast estimators for truncated and untruncated samples

Contrast	Estimator	var(D)	var($D\mid T$)
$\theta_B - \theta_A$	D_1	$2a\sigma^2(1-\rho)$	$\text{var}(D_1) + ab\sigma^2(1-\rho)^2$
$\lambda_B - \lambda_A$	D_2	$4a\sigma^2(1-\rho)$	$\text{var}(D_2) + ab\sigma^2(1-\rho)^2$
$\tau_B - \tau_A$	D_3	$a\sigma^2(1-\rho)/2$	$\text{var}(D_3)$

The biases are zero if $\eta = 0$, or, respectively for $D_1 \mid T$, $D_2 \mid T$ and $D_3 \mid T$, if $\rho_{13} = 1$, $\rho_{12} - \rho_{13} + \rho_{14} = 1$, and $\rho_{14} = \rho_{12}$.

In our original paper, we warned that lack of balance with respect to a truncated baseline measurement could lead to $c_1 \neq c_2$. Senn argues against this, saying that the analysis is conditional on the given randomization. We agreed with this statistical argument:[3] it is true that if we perform many such randomizations, we will not have to concern ourselves with lack of balance in the sense that the effect of lack of balance will average out over the long run. We called this 'statistical sophistry' because if a result is inaccurate, it matters little to the *scientist* whether the inaccuracy is due to lack of validity or lack of efficiency. On a given randomization, the lack of balance with respect to a truncated baseline measurement can easily create the impression of a treatment effect or the lack thereof. In other words, regression toward the mean could be a phenomenon that must be considered in 2×2 crossover designs with baseline measurements. In particular, the contrast estimators may be biased, depending on the correlation structure between periods, if either t_1 and t_2 are unequal or if the variances ε_1^2 and ε_2^2 are unequal. We note that if the variances are unequal then t_1 and t_2 will also be unequal. Moreover, $t_1 \neq t_2$ if $c_1 \neq c_2$ or $\mu_1 \neq \mu_2$.

In the example of our paper we compared the variances of the contrast estimators in samples with truncated versus untruncated baselines. Although the variances tend to be smaller in the truncated case, this difference is not the primary issue when considering regression toward the mean in 2×2 crossover designs. Nevertheless, we readily admit that some of the formulae for variances were wrong and we correct the erroneous formulae in the following discussion. Here, we assume equal means ($\mu_1 = \mu_2$), variances ($\varepsilon_1^2 = \varepsilon_2^2$) and cutoff points ($c_1 = c_2$). Based on the untruncated baseline measurement, the variance of the estimator of the contrast for first-order carryover effects, $\theta_B - \theta_A$, is

$$\text{var}(D_1) = 2a\sigma^2(1 - \rho_{13}),$$

where $a = (1/n_1 + 1/n_2)$. Conditional on the baseline measurements being truncated, the variance is

$$\text{var}(D_1 \mid T) = \text{var}(D_1) + ab\sigma^2(1 - \rho_{13})^2,$$

where $b = t(z - t)$, $z = (c - \mu)/\sigma$, $t = [\phi(z)/(1 - \Phi(z))]$, $\phi(z)$ and $\Phi(z)$ are the standard normal density and distribution functions, respectively, and $\mu = E(y_{ij})$ for all i, j. Also, the untruncated and truncated variances of the second-order carryover and treatment contrasts are, respectively

$$\text{var}(D_2) = 2a\sigma^2(2 - \rho_{12} + \rho_{13} - \rho_{14} - \rho_{23} + \rho_{24} - \rho_{34}),$$

$$\text{var}(D_2 \mid T) = \text{var}(D_2) + ab\sigma^2[(1 + \rho_{12}^2 + \rho_{13}^2 + \rho_{14}^2) - 2(\rho_{12} - \rho_{13} + \rho_{14} + \rho_{12}\rho_{13} - \rho_{12}\rho_{14} + \rho_{13}\rho_{14})]$$

and

$$\text{var}(D_3) = \frac{a}{2}\sigma^2(1 - \rho_{24}),$$

$$\text{var}(D_3 \mid T) = \text{var}(D_3) + \frac{a}{4}[b\sigma^2(\rho_{14}^2 + \rho_{12}^2 - 2\rho_{12}\rho_{14})].$$

If the correlations are uniformly equal to ρ, these variances reduce to those shown in Table I, which should replace the corresponding expressions in Table VII of our paper. Thus, $\text{var}(D_1 \mid T)$ is less than $\text{var}(D_1)$ and $\text{var}(D_2 \mid T)$ is less than $\text{var}(D_2)$ if $\rho > 0$ and the truncation is in the upper tail of the distribution. $\text{Var}(D_3 \mid T) = \text{var}(D_3)$ regardless of the value of ρ; it is this variance that Senn discusses in his letters. We thus agree that if the correlation structure is constant across time periods, $\text{var}(D_3)$ is the same irrespective of truncation on the baseline measurement. Conversely, if $\rho_{12} \neq \rho_{14}$, this would not be the case. Because correlations tend to decrease as the interval of time between two observations on the same variate increases, the conditional variance of D_3 may be expected to be smaller in many practical applications.

Finally, as we wrote in our original paper:

> 'Because we compare treatments within persons and we take more than one observation per person in a crossover trial, we often use small samples in these studies. In small samples we have an increased chance that in a given randomization one of the groups will tend to have values for the baseline variable more extreme than the other group. If the groups are not well balanced in this regard, the regression toward the mean will not be the same in the two groups and this could obscure the direct treatment contrast estimate ($\tau_B - \tau_A$) and the corresponding statistical test.'

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